International Journal of Data Science and Analytics (2026) 22:291
https://doi.org/10.1007/s41060-026-01261-z

REVIEW

Machine learning for wind and solar PV energy production forecasting: a systematic literature review

Dani Mahaini[1] · Marcos R. Machado[2] · João Luiz Rebelo Moreira[1]

Received: 30 March 2026 / Accepted: 7 August 2026
© The Author(s) 2026

Abstract

The growing share of renewable energy, especially solar photovoltaic (PV) and wind, increases the need for reliable power production forecasts beyond the very short term. Medium-term horizons (here defined as forecasts of at least 24 h ahead) are important for planning tasks such as market bidding, reserve sizing, and operational scheduling, but research in this area is diverse and difficult to compare due to differences in horizons, datasets, and evaluation settings. This paper presents a systematic literature review of machine learning approaches for PV and wind production forecasting with horizons $\geq$ 24 h. Following PRISMA guidelines, a Scopus search using a broad query (2978 results), followed by a refined query (69 results) and a three-stage screening process (title, abstract, and full text), resulted in 31 included studies. We extract and synthesize evidence on (i) forecast horizons and setups, (ii) model families and hybrid designs, (iii) data sources and the use of measured versus forecast weather inputs, (iv) uncertainty handling (e.g., quantiles, intervals, and scenarios), and (v) links to decision-making contexts. In addition, we propose a taxonomy to classify studies along these dimensions and support comparison across different evaluation settings. The results show that most studies focus on day-ahead and week-ahead evaluation, while 2-week and longer horizons are less common. Hybrid and ensemble approaches are widespread, while transformer-family methods appear in a smaller but visible subset. Most papers rely on measured production data, whereas fewer explicitly use forecast weather (e.g., Numerical Weather Prediction) inputs, even though weather-forecast uncertainty strongly influences multi-day performance in realistic settings. Only a limited number of studies connect forecasts to operational decisions, indicating a gap between accuracy-focused evaluation and decision value. The review highlights the need for more consistent benchmarking, more realistic weather-input evaluation (and, where relevant, other operational exogenous inputs such as curtailment), and more standard uncertainty reporting for medium-term renewable forecasting.

Keywords Renewable energy forecasting · Medium-term forecasting · Solar photovoltaic (PV) power · Wind power · Machine learning

1 Introduction

In the European Union, renewable energy sources accounted for 47.5% of gross electricity consumption in 2024 [25]. The increasing share of renewable energy, especially solar photovoltaic (PV) and wind, makes power systems more dependent on good production forecasts for safe and economic operation [19, 55]. Power production forecasts support daily and weekly planning tasks such as unit commitment, reserve procurement, maintenance planning, and market bidding [19, 51]. When forecast errors are large, grid operators and plant owners may face higher balancing costs, curtailment, or even reliability risks [19]. This is why forecasting research has moved from simple short-term point forecasts ($<$ 24 h ahead) toward more realistic multi-step and medium-term settings,

✉ Marcos R. Machado
m.r.machado@utwente.nl

Dani Mahaini
d.mahaini@student.utwente.nl

João Luiz Rebelo Moreira
j.luizrebelomoreira@utwente.nl

[1] Faculty of Electrical Engineering, Mathematics, and Computer Science, University of Twente, 7500 AE Enschede, The Netherlands

[2] Faculty of Behavioral, Management and Social Sciences, Department of High-Tech Business and Entrepreneurship, University of Twente, 7500 AE Enschede, The Netherlands

Published online: 03 September 2026

and toward methods that can quantify uncertainty [51]. As in practice, forecast errors can create balancing costs, for instance, wind forecast errors in Belgium have been estimated to incur roughly 1–4 €/MWh of wind energy produced [24].

Recent studies show that medium-term horizons (in this review defined as $\geq$ 24 h ahead) are actively examined for both PV and wind using machine learning (ML). For PV, many works focus on multi-horizon prediction and compare sequence-based time-series models and hybrid architectures. For example, Seman et al.[58] propose a hybrid deep learning forecasting model that builds on top of a bidirectional LSTM with additional components to better capture temporal patterns, and evaluate forecasts from 1 to 96 h ahead. Other approaches explore transformer-family models and stacking strategies to forecast renewable power generation (PV/wind) at horizons such as 24–168 h ahead [40]. Realistic operational inputs are also becoming more important: Polasek and Čadík [55] explicitly study forecasting under high-uncertainty weather forecasts and report performance across multiple forecast ages, highlighting that weather uncertainty can create a noticeable performance gap compared to ideal (error-free) inputs. In addition, preprocessing choices can strongly affect medium-term accuracy, Bergman et al. [8] shows this by comparing missing data and outlier handling strategies for multi-day PV forecasting.

For wind, medium-term forecasting is also commonly addressed with hybrid and ensemble strategies. Liao et al.[42] combine a decomposition-based linear component with LightGBM and Kalman filtering for 7-day forecasting, while Ekinci and Ozturk [21] compare multiple ML algorithms over short-, medium-, and long-term horizons for wind farm production forecasting. Some works aim to unify forecasting across renewable sources and across longer horizons by combining decomposition and neural network models [44]. Beyond studies that focus solely on predictive performance (e.g., minimizing mean absolute error (MAE)/root mean square error (RMSE)), other studies connect forecasting to decision-making under uncertainty. For instance, Osifeko & Munda [51] integrate ML-based forecasts with uncertainty quantification and scenario-based stochastic optimization for dispatch decisions, illustrating that forecast uncertainty is not only a modeling issue but also an operational one.

While there are now numerous studies proposing different model families, the literature is still fragmented across different horizons, datasets, weather-input assumptions, and evaluation protocols. This makes it hard for researchers and practitioners to compare results and to understand which design choices matter most in medium-term settings. In particular, three recurring challenges appear across the recent papers:

1. The increasing dependence on uncertain weather forecasts and the need to model that uncertainty [51, 55];
2. The lack of consistent multi-horizon benchmarking across model families and sites [38, 58];
3. Sensitivity to data preparation decisions and missing data handling, which can change conclusions about model quality [8, 37].

Therefore, this systematic literature review (SLR) synthesizes recent ML approaches for PV and wind power forecasting with horizons of at least 24 h ahead. Specifically, it summarizes which forecasting horizons and experimental setups are studied, which model families and hybrid combinations are used, and which data sources and weather-related features are assumed.

Based on this synthesis, the review aims to identify gaps and gives directions for future research, especially on uncertainty-aware forecasting and decision-oriented evaluation.

The review seeks to answer the following research questions:

- **RQ1:** What machine learning methods are used for day-ahead and multi-day solar PV and wind energy production forecasting?
- **RQ2:** How do forecast error metrics (e.g., MAE and RMSE) change when the forecast horizon increases?
- **RQ3:** How do different data sources and exogenous variables (e.g., measured SCADA data, NWP-based weather forecasts, and operational variables such as curtailment) influence forecast performance and evaluation results?
- **RQ4:** How do studies handle forecast uncertainty, i.e., the range of possible values around a forecast rather than one single number?
- **RQ5:** How are energy production forecasts connected to decision-making contexts (e.g., dispatch, planning, and operational use)?

This review contributes scientifically by synthesizing different medium-term PV and wind forecasting approaches into a structured comparison of horizons, model families, input assumptions, and evaluation practices, highlighting where results are and are not comparable. Practically, it supports operational stakeholders by translating these insights into guidance on data choices (measured vs. forecast weather), model selection, and uncertainty-aware forecasting for day-ahead to multi-day planning.

The remainder of this paper is structured as follows: Section 2 explains the methodology of the systematic literature review, including the search strategy, screening steps, and the data extraction process. Section 3 presents the descriptive results and main trends in the selected studies, such as forecast horizons, data sources, and model families. Next, Sect. 4

introduces a taxonomy that organizes the reviewed literature along key dimensions (horizon, inputs, model family, forecast output, and decision linkage), to support interpretation and comparison of studies. Subsequently, Sect. 5 discusses these findings in relation to the research questions, highlights key insights and limitations, and identifies gaps for future research. Finally, Sect. 6 concludes the paper by summarizing the main contributions of this review and providing recommendations for further work.

2 Methodology

This section explains how the systematic literature review (SLR) was conducted. It describes the database and search strategy, the screening procedure, and the data extraction approach. The literature review follows PRISMA framework [52] and the methodology was inspired by Fraccascia et al. [28] and yu et al. [71] papers.

2.1 Search strategy

The literature search was conducted using **Scopus**.[1] Scopus was selected because it indexes a large body of peer-reviewed literature in energy systems and ML and provides consistent bibliographic metadata (e.g., abstracts, keywords, and citation information) and advanced, reproducible search, and filtering options. These features support screening and data extraction while ensuring broad coverage of relevant papers. The search strategy was designed to identify studies that apply ML to renewable energy production forecasting, with a focus on medium-term horizons.

Two search strings were used. The first query was intentionally broad to understand the wider literature landscape, while the second query was refined to better match the scope of this review. The search took place at 9th of January 2026.
Search string 1 (broad query) The first search string included general renewable energy terms and a wide set of forecasting and ML keywords. This broad search returned **2978** results:

```
TITLE-ABS-KEY(
("renewable energy" OR solar OR wind
OR "green energy")
AND
("medium-term" OR "medium term" OR
"multi-step" OR "long-term" OR
"long term" OR "7-day" OR "7 day" OR
"14-day" OR "14 day")
AND
("machine learning" OR "ML" OR "deep
learning" OR "DL" LSTM OR XGBoost OR
"neural network" OR "NN")
AND
(forecast* OR predict*)
)
```

An initial inspection of these results showed that many papers were out of scope, for example because they focused on electricity price forecasting, demand/load forecasting instead of production, very short-term horizons, or renewable topics beyond wind and solar.
Search string 2 (refined query) Based on the observations from the broad query, a refined search string was created to narrow down the results to studies that are more likely to match this review. The refined query focused specifically on *solar and wind*, *energy production / power generation*, and multi-day horizon keywords, and it used a slightly more targeted set of ML terms. This refined search returned **69** publications, which were used as the starting set for systematic screening:

```
TITLE-ABS-KEY(
(solar OR wind)
AND
("energy production" OR "power
generation")
AND
("medium-term" OR "multi-day" OR
"7-day" OR "14-day" OR "two-week")
AND
("machine learning" OR "deep
learning" OR LSTM OR "gradient
boosting")
AND
(forecast* OR predict*)
)
```

2.2 Screening and eligibility criteria

The 69 publications were screened in three stages: title screening (Stage A), abstract screening (Stage B), and full-text screening (Stage C). The complete screening log table can be found in the Appendix 3. Additionally, Table 1 summarizes the inclusion and exclusion criteria.
Stage A: Title screening In Stage A, papers were excluded when the title indicated that the paper is not relevant to renewable production forecasting using ML. Typical exclusion reasons were: (i) demand/load forecasting instead of production, (ii) electricity price forecasting, (iii) no forecasting model (e.g., purely conceptual papers), (iv) focus only on short-term horizons ($\leq$ 24 h), or (v) unrelated energy domains.
Stage B: Abstract screening In Stage B, abstracts were screened to ensure that the study includes: (i) a ML model,

[1] https://www.scopus.com/home.uri.

Table 1 Inclusion and exclusion criteria used during screening

Stage	Inclusion indicators (must hold)	Typical exclusion indicators (examples)
A: Title	The title indicates renewable energy production forecasting using ML The topic appears relevant to PV and/or wind production forecasting	Forecasting target is demand/load rather than production Electricity price forecasting No forecasting model (e.g., purely conceptual / commentary) Focus only on short-term horizons ($<$ 24 h) Unrelated energy domains (not PV/wind production forecasting)
B: Abstract	Uses at least one ML model. Forecasts renewable energy production (PV or wind) Reports a multi-hour or multi-day horizon consistent with this review (i.e., $\geq$ 24 h ahead)	Does not use ML/ (e.g., purely physical model without ML component) Target variable not renewable energy power production (e.g., irradiance or wind speed forecasting) Horizon not aligned with the review scope (only intra-day/$<$ 24 h)
C: Full text	Confirms the study actually forecasts PV/wind power production Meets the horizon requirement ($\geq$ 24 h ahead). Provides sufficient methodological detail for synthesis (data, features, model)	Forecasts only meteorological proxies (e.g., wind speed, irradiance) rather than production output. Does not meet the horizon requirement upon full-text inspection. Insufficient methodological detail (e.g., missing data description, model setup)

(ii) a renewable production target (PV or wind power/energy) and (iii) a multi-hour or multi-day forecasting horizon consistent with this review.

Stage C: Full-text screening In Stage C, the full text of the remaining papers was read to confirm eligibility and extract details for synthesis. At this stage, papers were excluded if they did not actually forecast renewable production (e.g., forecasting only wind speed or solar irradiation), did not meet the horizon requirement, or did not provide sufficient methodological detail.

After completing the screening process, **38** papers were excluded and **31** papers remained for synthesis. Figure 1 summarizes the step-by-step selection process and the paper counts at each stage.

2.3 Data extraction and synthesis

For each included paper, information was extracted using a structured template with five fields: (1) author and year, (2) paper title, (3) forecast horizon(s), (4) models used, and (5) context. The context field includes dataset type (e.g., SCADA or simulated data), location, time resolution, input variables, and evaluation metrics. This template supports consistent comparison across papers. The complete data extraction can be found in the Appendix, in Table 2.

The synthesis was performed in two parts: First, a descriptive synthesis summarizes patterns across the selected studies, such as the distribution of forecast horizons and the most common model families. Second, a thematic synthesis groups findings related to uncertainty handling, the use of weather forecasts versus measured inputs, and the connection to decision-making contexts. These results are presented in Sect. 3 and discussed in Sect. 5.

3 Results

3.1 Quantitative results

This subsection provides a quantitative overview of the selected papers in this literature review. First, Fig. 2 reports the yearly distribution of publications from the refined Scopus query ($n = 69$). Second, two bibliometric maps were created using VOSviewer[2] based on the selected studies after screening to illustrate (i) the thematic structure of the included set and (ii) how closely the included papers are connected through shared references.

3.1.1 Publication trend (refined query)

Figure 2 shows an increasing number of publications over time for the refined query set. This trend indicates that

[2] https://www.vosviewer.com/.

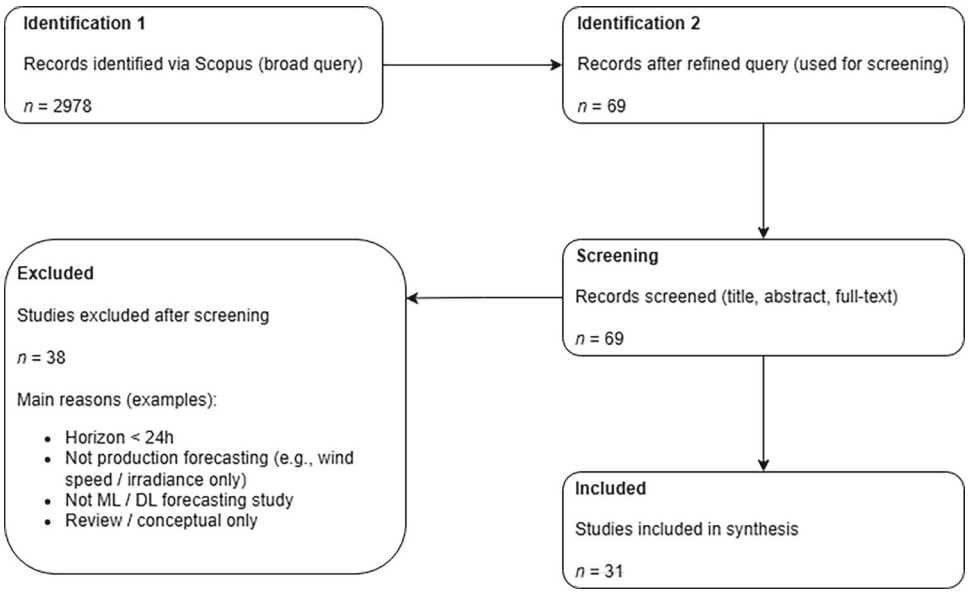


Fig. 1 PRISMA-like flow diagram of the literature selection process

medium-term wind and PV forecasting with ML has attracted growing attention in recent years.

3.1.2 Keyword co-occurrence

Figure 3 presents the keyword co-occurrence network for the included studies. The map shows several connected clusters. One cluster contains method-related terms such as *deep learning*, *power forecasting*, and *uncertainty analysis*, while another cluster includes domain terms related to PV and renewable energy (e.g., *photovoltaics*, *renewable energy*). A further cluster connects terms related to wind forecasting and weather-related modeling (e.g., *weather forecasting*, *wind power*, *long short-term memory*). Overall, the network suggests that the selected literature covers both domain-specific streams (wind and PV) and method streams (ML and uncertainty).

3.1.3 Bibliographic coupling

Figure 4 shows a bibliographic coupling map for the included studies, where links represent shared references. Most studies form one main connected group, which suggests that a large part of the selected literature builds on a common reference base. At the same time, a smaller number of papers appear weakly connected or more isolated, indicating that they draw on different background literature and may represent more specialized application settings.

3.2 Qualitative results

This subsection presents the review's results by summarizing the included studies (Table 2) in terms of forecast horizons, model families, data/input assumptions, uncertainty handling, and operational relevance. The results are structured to support the research questions introduced in Sect. 1. First, the distribution of forecast horizons is summarized (RQ2) and visualized in Fig. 5. Second, the main model families are summarized (RQ1) and shown in Fig. 6. Third, the most common data sources and input settings are described (RQ3) and visualized in Fig. 7. Finally, this section reports how uncertainty is handled (RQ4) and how forecasts are connected to decision-making contexts (RQ5).

3.2.1 Forecast horizons and forecasting setups

Across the reviewed studies, medium-term power production forecasting is operationalized in different ways. A large share of papers evaluates day-ahead forecasting ($\approx$ 24 h) and multi-day horizons up to four days (e.g., Seman et al. [58]; Bergman et al. [8]; Markovics and Mayer [45]). A second group explicitly targets week-ahead horizons (e.g., Osifeko and Munda [51]; Liao et al. [42]; Alam et al. [3]). Fewer studies evaluate two-week horizons (e.g., Alhassan et al. [4]; Wang et al. [66]; Guo et al. [29]), Alamends evaluation to three to four weeks or longer (e.g., Mirza et al. [47]; Malhan & Mittal [44]; Fahim et al. [26]).

Figure 5 summarizes how often each horizon category appears across the included papers. Overall, the literature is strongest for day-ahead, 2–4 day, and week-ahead evaluation, while explicit 2-week and multi-week evaluation remains less common. Several studies also adopt multi-horizon (multi-timescale) setups to forecast PV or wind power output at multiple lead times within one framework, for example by evaluating 1-day, 3-day, and 7-day ahead forecastss [38] or by combining short-, medium-, and long-term

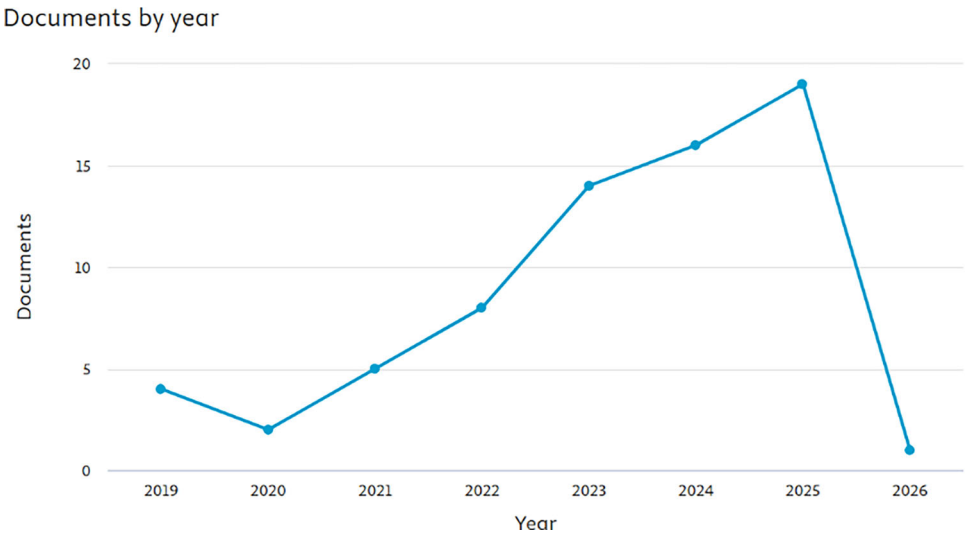


Fig. 2 Documents by year for the refined search results ($n = 69$) before in-depth screening

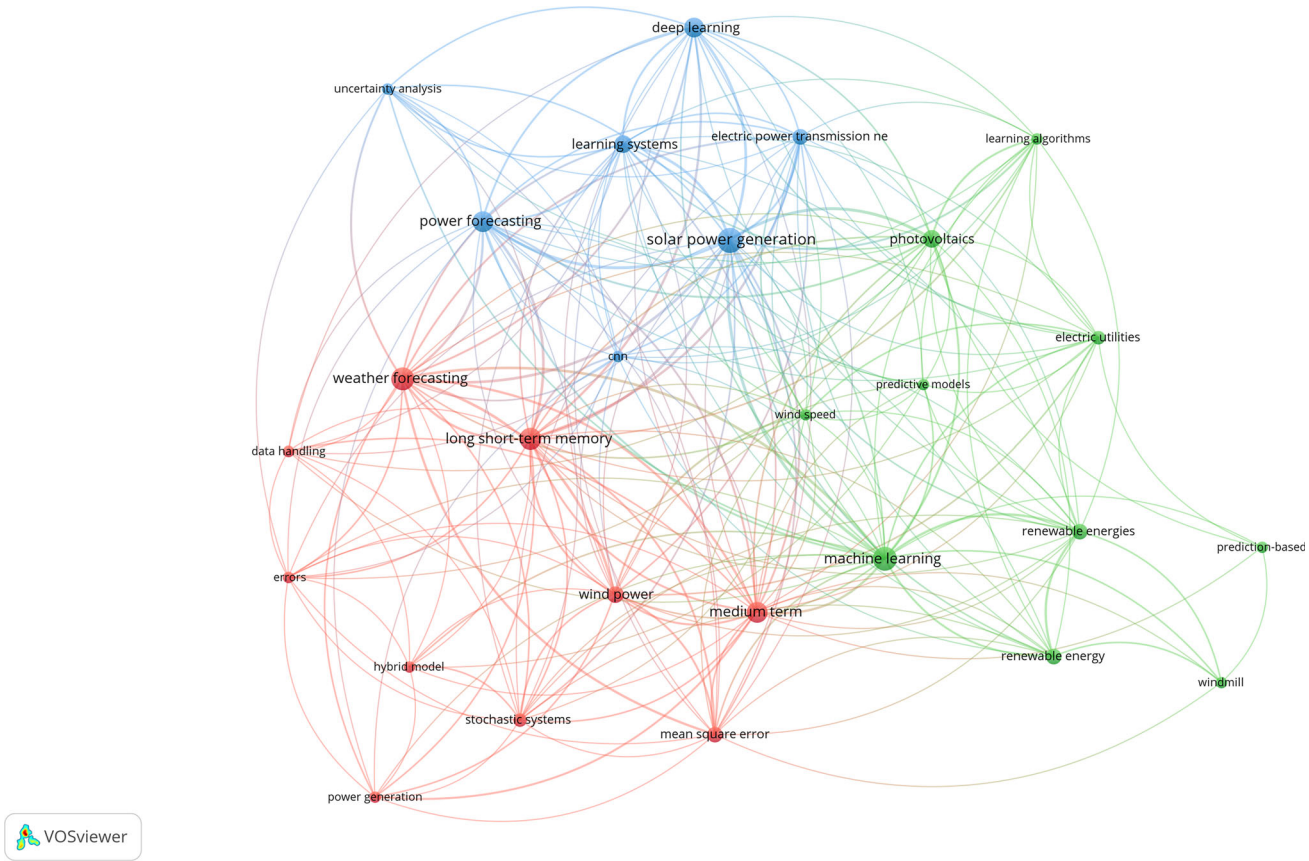


Fig. 3 VOSviewer keyword co-occurrence network for the included studies ($n = 31$)

horizons in a single study [12]. This is practically relevant because different lead times support different operational decisions (e.g., day-ahead bidding versus multi-day maintenance and reserve planning) [19, 51]. In addition, models for longer horizons typically rely more on forecast-based exogenous inputs (e.g., NWP weather forecasts and their update cycles) [41], whereas shorter horizons can benefit more from recent measurements and lagged production signals, which affects how studies design and evaluate their inputs [58].

3.2.2 Model families and learning approaches

Across the included studies, the forecasting task is almost always formulated as a regression problem, where the tar-

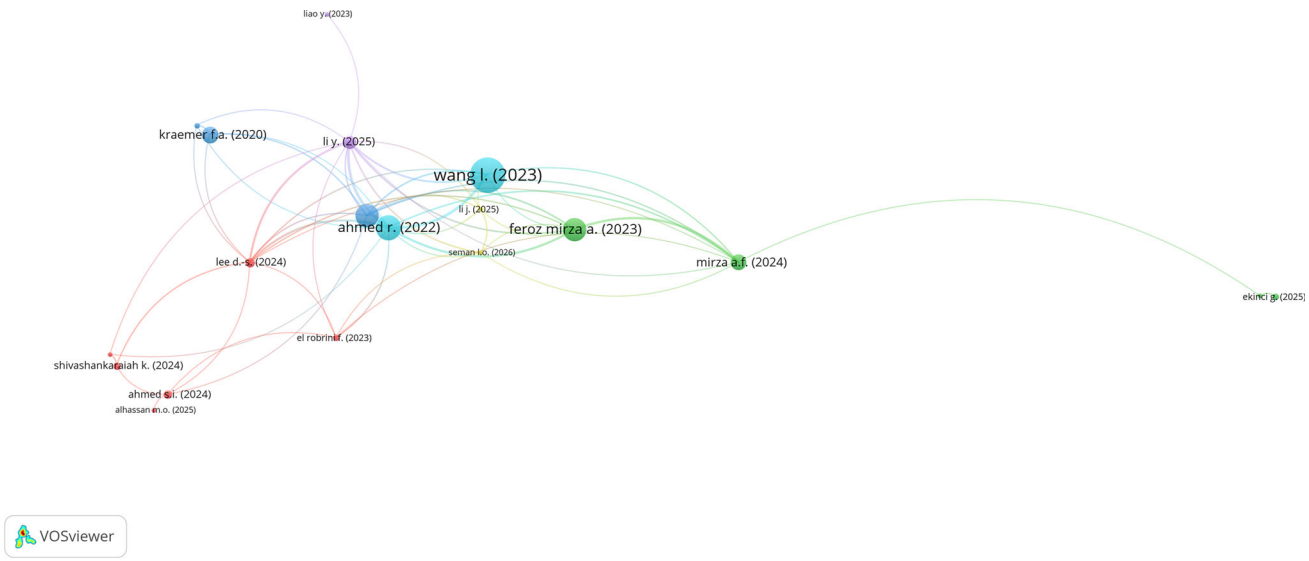


Fig. 4 VOSviewer bibliographic coupling map (documents) for the included studies ($n = 31$); links represent shared references

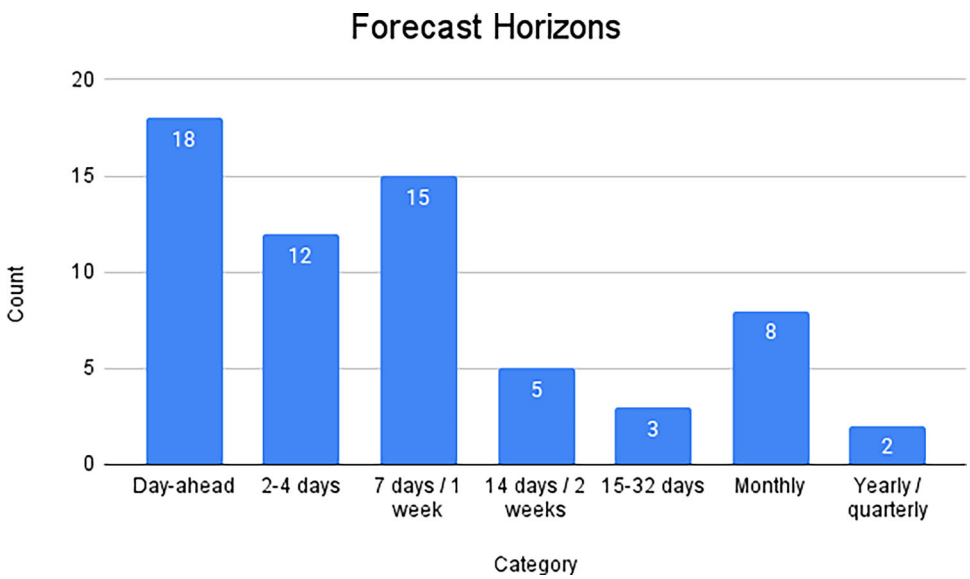


Fig. 5 Distribution of forecast horizons across the included studies ($n = 31$). Papers may appear in multiple bins

get is continuous renewable power production (PV or wind) rather than a discrete class. The dominant goal is therefore predictive accuracy (MAE/RMSE and related metrics), not classification.

In terms of purpose, most papers are predictive rather than prescriptive, they propose forecasting models and evaluate error metrics, but they do not explicitly connect the forecasts into an optimization or control decision. An exception is work that links forecasts to operational planning, for example, scenario-based dispatch and scheduling under uncertainty [51].

For clarity, the included studies can be grouped into the following model families and learning approaches:

- Tree-based ensemble learning. Tree ensembles such as Random Forest, XGBoost, and LightGBM appear frequently, either as standalone forecasters or as strong baselines [8, 41, 45]. These models are popular because they train efficiently, capture nonlinear feature interactions, and work well with engineered inputs (lags and weather variables). They also fit operational settings where retraining speed and robustness matter, as shown in full real-life wind prediction studies using gradient boosting with forecast inputs [41].
- Recurrent neural networks. LSTM- and GRU-based models are commonly used for multi-step forecasting and benchmarking [4, 58]. Their strength is sequence modeling using recent history (lagged production and meteorological measurements). Additionally, some stud-

ies use LSTM ensembles with adaptive weighting across horizons [1].

- Transformer-family and attention-based models. Attention-based models appear in a smaller but visible subset of the included studies [36, 47]. These approaches are typically motivated by the need to model longer dependencies and multi-horizon outputs. In PV forecasting, transformer-related components also appear as part of hybrid stacking pipelines rather than as a single end-to-end model [40]. In general, compared to tree ensembles and LSTMs, transformer-family approaches seem to be an emerging direction rather than the dominant standard in the current medium-term renewable energy production forecasting literature.
- Hybrid, decomposition, and stacking strategies. Hybridization is the most recurring design pattern across the reviewed papers. Many studies combine decomposition or signal processing with ML predictors, or stack multiple learners using a meta-model. Examples include wavelet or frequency-domain decompositions combined with deep models [38, 66] and hybrid pipelines that integrate linear components, boosting, and post-processing (e.g., Kalman filtering) [42]. These designs are often motivated by medium-term challenges, such as seasonality and rising uncertainty as the production forecast horizon increases.

A pattern identified across the reviewed papers is that the modeling approach often depends on how far ahead the forecast is made. When the horizon is closer, studies often use recent measurements and lagged production values as key inputs [1, 58]. When the horizon is longer (multi-day), studies more often rely on weather forecast inputs such as NWP and their updates, and they frequently use hybrid designs or attention-based components to keep the multi-day predictions stable [38, 41, 55]. Overall, this suggests that forecasting performance depends not only on the chosen model family, but also on which inputs are available and used at the target horizon.

Figure 6 summarizes the main model families used across the included studies. Because many papers combine components, the categories are not mutually exclusive.

3.2.3 Data sources and weather input assumptions

The reviewed papers use different data sources and input assumptions, which strongly affect realism and observed performance.

- Measured production (SCADA/plant monitoring). Most studies use measured wind or PV production from real plants, often combined with local meteorological variables [4, 8, 12]. Additionally, Bergman et al. [8] evaluate their models using the publicly available Desert Knowledge Australia Solar Centre (DKASC) dataset.[3] These datasets reflect operational noise and seasonal variability and are therefore common in applied forecasting studies.
- Measured production with forecast weather (NWP or public forecasts). A smaller subset explicitly uses weather forecast products as model inputs, which is important because forecast errors increase with horizon length. For example, Li et al. [41] evaluates wind forecasting in a full real-life setting using ECMWF-type forecast inputs, and Markovics and Mayer [45] studies how the training window interacts with NWP-based PV forecasting. Polasek and Čadík [55] goes further by training PV models on realistic high-uncertainty forecasts and by considering forecast "age" as part of the learning problem, using the publicly available SolarDB dataset.[4]
- Simulated production data. A smaller group relies on simulated PV output (e.g., PVWatts/NSRDB-type simulations)[5] to enable controlled benchmarking [2, 6]. While simulation can be useful for comparing architectures, it may not capture operational effects such as curtailment, sensor issues, or imperfect forecasts.

Overall, Fig. 7 shows that measured production datasets dominate, while fewer studies explicitly use forecast weather (NWP) inputs. This is important because multi-day forecasting performance is strongly affected by weather-forecast uncertainty [55], and operational wind forecasting at longer horizons commonly depends on NWP forecast products [41].

3.2.4 Uncertainty handling and probabilistic outputs

Most studies still report deterministic point forecasts evaluated with metrics such as MAE and RMSE. However, several papers include uncertainty-aware outputs such as quantiles, prediction intervals, or scenarios. For example, Osifeko and Munda [51] combines forecasting with uncertainty quantification and scenario generation for planning, while Kumar and Picerno [36] applies quantile regression within a multi-horizon transformer framework. In addition, prediction-interval or reliability-oriented evaluation appears in some PV studies [2, 66]. These examples show that uncertainty-aware forecasting is feasible and used; however, it is not yet a default choice: most reviewed studies still focus on point forecasts only, and uncertainty is typically included as an additional analysis rather than reported as the main forecasting output.

[3] https://dkasolarcentre.com.au/.

[4] https://cphoto.fit.vutbr.cz/solar/.

[5] PVWatts: https://pvwatts.nrel.gov/.; NSRDB: https://nsrdb.nrel.gov/.

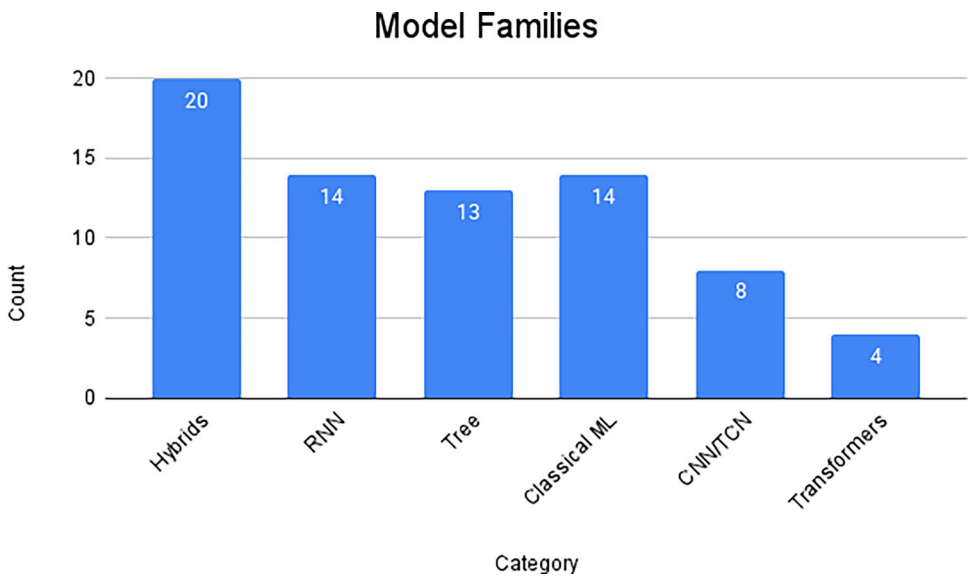


Fig. 6 Distribution of model families across the included studies ($n = 31$). Papers may appear in multiple bins

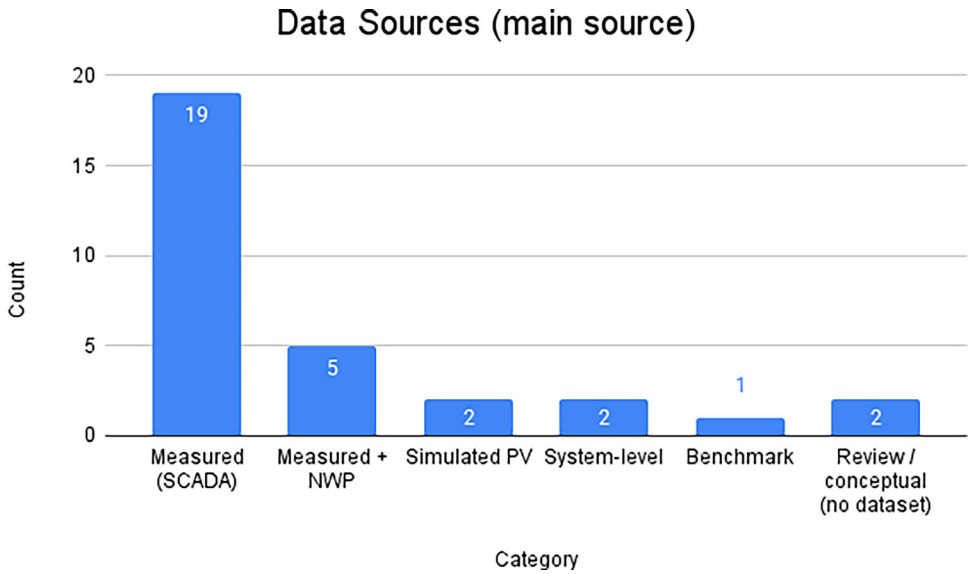


Fig. 7 Distribution of data sources across the included studies ($n = 31$)

3.2.5 Decision-making context and operational relevance

Most reviewed studies focus primarily on predictive accuracy and comparison across models. Only a smaller subset links forecasts directly to an operational decision context. A clear example is Osifeko & Munda [51], where probabilistic forecasts are translated into scenarios for dispatch optimization under uncertainty. Other decision-oriented contexts include energy management for autonomous systems [35] and predictive modeling in a wind turbine digital twin setting [26]. In addition, review-oriented papers emphasize that forecasting value is often realized only when integrated into grid or market decisions [19, 22].

4 Taxonomy

The results in Sect. 3 show that medium-term renewable energy production forecasting studies differ not only in model choice, but also in how they define the horizon, which inputs they assume, which type of forecast they output and whether the forecast is connected to an operational decision. To make these differences easier to compare, this section proposes a simple taxonomy that organizes the reviewed literature along five practical dimensions. The goal is not to rank methods, but to provide a clear map of "what kinds of studies exist" and how they relate to each other.

Figure 8 summarizes the taxonomy as a forecasting pipeline. The first dimension is the forecast horizon. Many papers evaluate day-ahead settings (24 h), while others target multi-day or week-ahead horizons, and some use multi-

horizon setups where several lead times are predicted in one framework [51, 58]. This dimension matters because the difficulty of the task increases with lead time, and input availability also changes.

The second dimension is the inputs and data assumptions. A large share of studies rely on measured production (and often measured meteorological variables) [8, 12]. A smaller subset explicitly uses forecast weather products (e.g., NWP or public weather forecasts), which increases realism for multi-day horizons but also introduces additional uncertainty [45, 55]. Finally, some studies use simulated PV output (e.g., PVWatts/NSRDB-type simulations), which can support controlled benchmarking but may not reflect operational effects such as curtailment and imperfect forecasts [2, 6].

The third dimension is the model family. Based on RQ1, the most frequent families include tree ensembles (RF/XGBoost/LightGBM), recurrent neural networks (LSTM/GRU), transformer/attention-based approaches, and hybrid designs that combine preprocessing or decomposition with ML learners [8, 58]. Importantly, many studies use more than one family in a single pipeline, so the categories are not mutually exclusive [42, 66].

The fourth dimension is the forecast output. Most studies produce point forecasts, but some provide uncertainty-aware outputs such as quantiles or prediction intervals, and a smaller subset generates scenarios [51]. This dimension is closely related to how the forecasts can be used: Uncertainty-aware outputs are often more informative when decisions must be made under risk.

The final dimension is the use in decisions. Many papers remain accuracy-focused and evaluate models using error metrics only [38]. In contrast, a smaller group links forecasts to an operational context (for example, planning or dispatch), where forecast quality affects cost or reliability outcomes [35, 51]. This dimension is important for the overall relevance of the research, because the value of forecasting is ultimately realized when it supports real decision-making.

To summarize, the taxonomy provides a compact framework for interpreting the results: Studies can be described as combinations of (horizon, input realism, model family, output type, and decision linkage). This also helps to highlight gaps that are discussed in the next sections, for example, the limited number of studies that combine realistic forecast weather inputs with uncertainty-aware outputs and explicit operational evaluation.

5 Discussion

This section discusses the main findings of the review in relation to the research questions (RQ1–RQ5). Overall, the review findings indicate that medium-term forecasting research is active and expanding, but remains fragmented across horizons, datasets, and evaluation choices. Figures 3, 4 and 5 highlight three broad patterns: (i) medium-term horizons are studied most often up to 1 week, with fewer studies extending to 2 weeks or longer; (ii) hybrid and ensemble designs are very common, while transformer-style approaches appear less frequently; and (iii) measured production data dominates, while explicit use of forecast weather (NWP) inputs is less common, even though it strongly affects multi-day performance in realistic settings (e.g., Polasek and Čadík [55], Li et al. [41]).

5.1 RQ1: What ML methods are used for day-ahead and multi-day forecasting?

The reviewed studies show that there is no single dominant model type for medium-term PV and wind forecasting. Instead, many papers combine multiple components, which helps explain why "hybrids" appear as the most frequent model family in Fig. 6. Hybridization often takes the form of decomposition + learning, stacking, or adding a post-processing step. Examples include wavelet/frequency-domain ideas combined with deep learning [38, 66] and decomposition plus boosting and filtering [42]. A reasonable interpretation is that longer horizons amplify challenges such as non-stationary weather patterns, missing data, and seasonality, so authors rely on hybrids to handle different parts of the signal.

At the same time, the results confirm that classical and widely used families remain strong baselines. Tree-based ensembles (RF, XGBoost, and LightGBM) are widely used and are often competitive, especially when feature engineering is strong [8, 41, 45]. RNN-based models (LSTM/GRU) are still common and frequently used as benchmarks or core learners in multi-step settings [4, 58]. Transformer-family models appear in fewer papers, but their presence in medium-term work suggests an emerging direction rather than a settled best practice [36, 47]. Overall, the literature indicates that progress is driven more by combinations (preprocessing + model + evaluation design) than by a single architecture choice.

5.2 RQ2: How do forecast error metrics (e.g., MAE and RMSE) change when the horizon increases?

The horizon distribution in Fig. 5 supports a practical point: Most studies concentrate on day-ahead and week-ahead horizons, while 2-week and longer horizons are comparatively rare. This matters because medium-term performance is not only a function of the model but also of the inputs. As horizons increase, forecast errors typically grow due to weather uncertainty and reduced predictability, especially for PV and wind [19]. Several papers indirectly respond to this by using

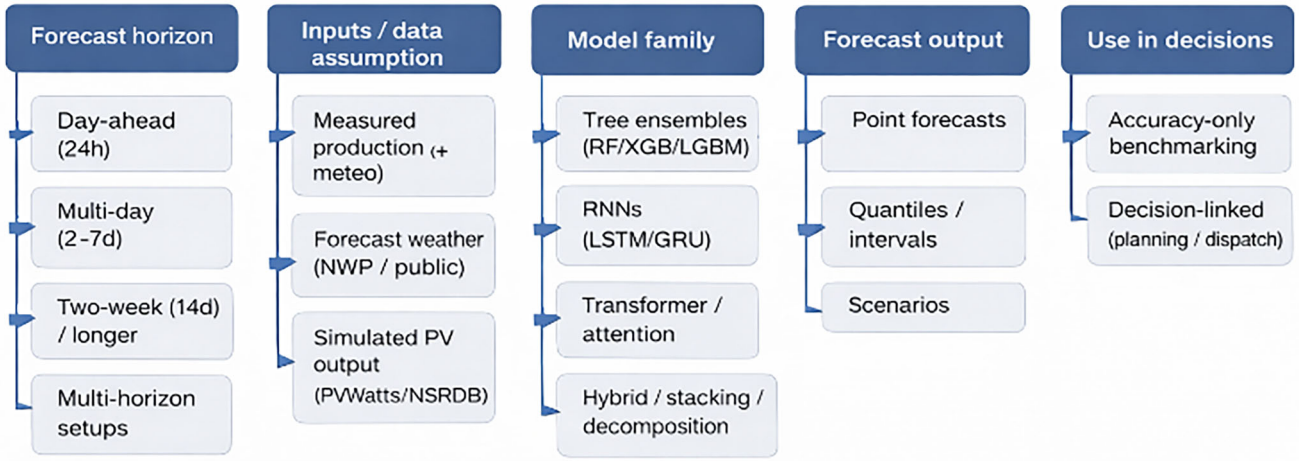


Fig. 8 Proposed taxonomy for classifying medium-term to long-term renewable energy production forecasting studies (horizon $\geq$ 24 h)

multi-horizon evaluation (e.g., 1-day, 3-day, and 7-day) and multi-timescale designs [12, 38].

However, cross-study comparisons of performance degradation associated with horizon length remain challenging because studies differ in datasets, time resolutions, training windows, and evaluation metrics. For example, some papers report a set of horizons but use different sampling frequencies or different validation strategies, which makes cross-paper comparison hard even when the horizon labels are similar. This suggests a clear research opportunity: medium-term forecasting would benefit from more consistent evaluation protocols (e.g., common horizon sets, rolling-window evaluation, multi-site testing, and clearer reporting of data preparation steps), as highlighted by preprocessing-focused work [8, 37].

5.3 RQ3: How do different data sources and exogenous variables influence outcomes?

The findings in Fig. 7 show that most papers rely on measured production data (often with measured meteorological variables), while fewer studies explicitly incorporate forecast weather (NWP) inputs or other operational exogenous variables (e.g., curtailment, maintenance information, or plant metadata). This difference is important for interpretation. Models trained and evaluated using measured historical weather can appear very accurate, but such evaluations may underestimate real operational error when future weather is unknown. In contrast, studies that explicitly use forecast weather and consider forecast uncertainty provide a more realistic view of medium-term performance.

The clearest evidence for this is work that focuses on forecasting under high-uncertainty weather forecasts. Polasek and Čadík [55] explicitly demonstrate that realistic forecast uncertainty can create a noticeable performance gap compared to idealized inputs. Similarly, operational wind forecasting studies often assume NWP forecasts as part of the input pipeline, because multi-day wind forecasting depends on the availability and quality of these forecast products [41]. From a practical perspective, this suggests that future medium-term forecasting research should more often report results under realistic weather forecast conditions, rather than only under measured-weather settings, especially when the intended use case is operational planning.

5.4 RQ4: How is forecast uncertainty handled?

The review indicates that uncertainty handling exists, but it is not yet standard across the medium-term forecasting literature. Most studies still focus on point forecasts and deterministic metrics (MAE and RMSE), while fewer studies provide probabilistic outputs (quantiles and intervals) or scenario-based representations. Nevertheless, several studies explicitly model forecast uncertainty by providing probabilistic outputs such as quantiles, prediction intervals, or scenario ensembles. For example, Osifeko and Munda [51] combine forecasting with uncertainty quantification and scenario generation, and Kumar and Picerno [36] apply quantile regression within a multi-horizon forecasting approach. In PV forecasting, interval-focused evaluation also appear [2, 66].

A key implication is that uncertainty handling should be matched to the decision context (RQ5). For operational planning, knowing the expected value is often not sufficient, risk-aware decisions aim to limit exposure to adverse outcomes (e.g., large under-forecasts that lead to imbalance costs or reserve activation, or over-forecasts that cause infeasible schedules, unnecessary curtailment, or inefficient storage). In such settings, practitioners need information about forecast uncertainty, such as prediction intervals, quantiles, or

scenario ensembles to support various actions associated with energy planning and decision making. This aligns with broader forecasting guidance in the wind forecasting literature that emphasizes uncertainty and operational value, not only accuracy [19]. Therefore, an important gap is that many medium-term studies still report strong point-forecast performance without discussing whether errors are systematic, whether uncertainty grows with horizon, or how prediction uncertainty should be communicated to users in decision-relevant terms (e.g., expected cost/benefit, risk buffers, or profit-impact for biddings on energy market).

5.5 RQ5: How are forecasts connected to decision-making contexts?

The review confirms that most papers treat forecasting as an accuracy optimization problem and stop at model comparison, while only a smaller subset directly links forecasts to decision-making. A clear example is Osifeko & Munda [51], where forecasts and uncertainty are translated into scenario sets for dispatch optimization. Other decision-oriented contexts exist, such as energy management for autonomous systems [35] or predictive modeling in wind turbine digital twins [26]. In addition, review-oriented work emphasizes that the value of forecasting is realized when forecasts are integrated into grid or market decisions [19, 22].

This gap suggests two practical directions. First, future research can evaluate models not only on MAE/RMSE but also on operational outcomes (e.g., cost, reserve sizing, imbalance risk, and curtailment reduction). Second, papers should be clearer about the target user (TSO/DSO, plant owner, and market trader) and the decision that the forecast supports. Without this, it is difficult to interpret what level of accuracy is "good enough" and whether a model improvement is meaningful in practice.

5.6 Additional observations

Beyond the stated research questions, two recurring observations emerged across the included studies that are particularly relevant for operational deployment: interpretability and computational reporting.

Across the reviewed literature, evaluation is still dominated by predictive error metrics, with limited attention to explainability beyond basic feature-level reporting. Interpretability is most visible in tree-based models, where studies commonly report feature importance or sensitivity-style analyses, whereas deep learning papers more often motivate architectural choices rather than explaining individual predictions. For example, Bergman et al. [8] show that preprocessing decisions can materially change performance, which reinforces the need for transparent reporting of data handling, and Seman et al. [58] include an additive component partly intended to improve interpretability relative to purely black-box models. However, more systematic XAI practices (e.g., local explanations, uncertainty-aware explanations, or stakeholder-oriented interpretation) are not consistently reported across studies. For decision support, clearer interpretability is valuable because practitioners often need to understand which inputs drive the forecast and under which conditions performance degrades. A practical recommendation is that future medium-term studies report interpretable drivers (e.g., weather variables and lag effects), document data preparation choices, and, where appropriate, include a lightweight explanation layer (e.g., SHAP for tree ensembles or feature attribution methods for neural models).

A second recurring observation concerns computational aspects such as training time, inference latency, and model update frequency. While most papers report accuracy metrics, computational cost is described less consistently, even though it can influence model choice in operational settings. Li et al. [41], for instance, motivate LightGBM partly on practical grounds, including faster retraining, which becomes relevant when models are updated frequently as new measurements arrive or as weather forecasts are refreshed. In practice, training time matters for pipelines that retrain daily or weekly, and inference time matters when producing forecasts for many assets or multiple horizons simultaneously. Yet, the reviewed studies often provide limited information (e.g., hardware specifications, training duration, and inference time) that would enable comparison of computational efficiency across model families. Reporting basic computational details alongside predictive performance would therefore improve reproducibility and help practitioners assess whether complex hybrid approaches offer a favorable accuracy–complexity trade-off.

5.7 Cross-cutting implications and gaps

Across RQ1–RQ5, three cross-cutting gaps appear.

1. *Lack of standardized benchmarking.* The studies vary widely in horizon sets, time resolutions, train-test splits, and reporting. This makes it hard to compare methods and to identify robust best practices. Even when papers study similar horizons, they often do so under different weather-input assumptions and different preprocessing choices [8, 37]. In addition, reporting practices are often not detailed enough to support reproducibility and interpretation, for example regarding data preparation decisions and the drivers behind model behavior.
2. *Limited realism regarding weather forecast uncertainty.* Figure 7 shows that explicit NWP/forecast weather usage is less common than measured-weather settings. However, the papers that do model realistic forecast uncer-

tainty show that it strongly affects multi-day performance [55]. This indicates that medium-term forecasting research should more often treat weather forecasts as uncertain inputs and report results accordingly.

3. *Weak connection between forecasts and value.* Only a subset of papers evaluates forecasts in a decision-oriented setting [51]. For many applications, the relevant question is not only "which model has lower RMSE" but also "which forecast improves planning outcomes". This is especially important at longer horizons, where uncertainty is high and decision value depends on risk sensitivity [19]. From an operational perspective, it also helps when studies explain why the model makes certain predictions (interpretability) and report practical details such as training time, inference time, and how often the model needs updating, because these factors affect whether a method can realistically be deployed and reproduced, not just how accurate it is.

Overall, the findings suggest that further progress in medium-term PV and wind forecasting will likely depend on using more realistic evaluation settings, such as forecast-based weather inputs and multi-site testing, together with more consistent benchmarking protocols across studies. In addition, the literature would benefit from stronger links between renewable energy generation forecasting outputs, uncertainty quantification, and the operational decisions these forecasts are meant to support.

6 Conclusion

This systematic literature review synthesized recent machine learning approaches for medium-term renewable energy production forecasting (defined in this review as horizons of at least 24 h ahead). Based on 31 selected studies, the review summarized the main forecast horizons, model families, data sources, and evaluation choices and discussed how current research addresses uncertainty and operational relevance. As well as introduced a simple taxonomy that groups studies by horizon, inputs, model type, output type, and operational use, to make comparisons across papers more transparent.

Overall, the results show that most studies focus on day-ahead and week-ahead settings, while explicit 2-week and longer horizons are less common. In terms of methods, hybrid approaches are widespread, often combining decomposition, feature engineering, and multiple learning models in one pipeline. In terms of data, measured production datasets dominate, while fewer studies explicitly use forecast weather (NWP) inputs. However, the studies that do consider realistic forecast inputs show that weather-forecast uncertainty can strongly influence multi-day forecasting performance [55]. Finally, only a smaller subset of papers connects forecasts directly to decision-making contexts such as planning or dispatch under uncertainty [51].

6.1 Implications of the study

This review has several implications for both researchers and practitioners. First, the dominance of hybrid pipelines suggests that medium-term performance is often driven by the full design (data preparation + features + model + evaluation), rather than by a single algorithm choice. This supports the continued use of strong baselines (e.g., boosting ensembles and LSTM-type models) while emphasizing careful preprocessing and feature construction [8, 37].

Second, the findings highlight that dataset realism matters for interpreting results. Studies evaluated with measured historical weather inputs can underestimate operational error, especially for multi-day horizons. Therefore, medium-term forecasting research should more often report results under realistic forecast weather conditions, including uncertainty and forecast-update effects [41, 55].

Third, uncertainty-aware forecasting remains underused relative to its practical value. For decision support, point forecasts alone are often not sufficient. Instead, probabilistic outputs such as quantiles, intervals, or scenarios can better support risk-aware planning [19, 51]. This suggests that future work should treat uncertainty not only as an add-on, but as a standard part of medium-term forecast design and evaluation.

Finally, the limited connection between forecasting accuracy and operational value suggests an important direction for applied research. Models could be evaluated not only by MAE/RMSE, but also by their impact on operational outcomes such as imbalance costs, reserve sizing, curtailment reduction, or planning reliability [19, 51].

6.2 Limitations and future recommendations

This review has several limitations. First, the search was conducted using Scopus and a specific set of keywords, so relevant papers indexed only in other databases or using different terminology may have been missed. Second, the included studies are heterogeneous in horizons, time resolution, validation strategy, and reported metrics, which limits strict cross-paper comparison. Third, the review is based on reported results in the papers; differences in implementation details, hyperparameter tuning effort, or unavailable preprocessing steps may influence the comparability of outcomes.

Based on the synthesis, the following recommendations are suggested for future research:

- More consistent benchmarking.
  Use clearer and more comparable evaluation protocols, such as rolling-window validation, shared horizon sets, and multi-site testing. This would improve comparability across model families and datasets.
- More realistic weather-input settings.
  Evaluate multi-day forecasts using forecast weather inputs (NWP or public forecasts) and explicitly report how forecast uncertainty affects performance [55].
- Stronger uncertainty reporting.
  Increase the use of probabilistic forecasting and report uncertainty growth over the horizon using quantiles, prediction intervals, or scenario representations [51].
- Decision-oriented evaluation.
  Connect forecasts to the planning task they support and evaluate models using decision-relevant metrics (e.g., cost or reliability impact), not only prediction error [19].

Appendices

Data extraction table

Screening log table

Table 2 Data extraction table of included studies (forecast horizon, models, and context)

Author	Paper title	Horizon (forecast)	Models used	Context
[58]	Multi-step short-term solar energy forecasting using Fourier-enhanced BiLSTM and neural additive models	1–96 h ahead (best at 6 h)	Hybrid DL: Fourier Neural Operator (FNO) + BiLSTM (seq2seq, attention, teacher forcing) + Neural Additive Model (NAM); baselines: RF, XGBoost, LightGBM, NGBoost, TabPFN, LSTM, Transformer	PV forecasting for a 5 MW solar plant (Brazil) using historical PV output + meteorological + temporal features; accuracy + interpretability
[7]	A Novel Scenario-Based Comparative Framework for Short- and Medium-Term Solar PV Power Forecasting Using Deep Learning Models	1-hour ahead to 1-month ahead	LSTM, GRU, 1D-CNN (comparative DL framework; repeated runs/early stopping)	250 kWp grid-connected PV at Dicle University (Turkey); 1 year of real-time PV + meteorological data (hourly aggregation)
[51]	Scenario-Based Stochastic Optimization for Renewable Integration Under Forecast Uncertainty: A South African Power System Case Study	7-day (168 h) operational horizon	Hybrid LSTM–XGBoost forecasts; uncertainty via MC dropout + quantile regression; scenario generation + reduction; two-stage stochastic MILP dispatch	South African case (Eskom-style system); wind/PV/CSP + demand with forecast uncertainty; evaluates dispatch impacts
[38]	Multi-timescale photovoltaic station power prediction based on Reformer model	1-day/3-day/7-day ahead	Proposed WTC-Reformer-KAN (wavelet conv + lightweight Reformer encoder + KAN output); baselines: LSTM/GRU/BiLSTM/BiGRU/Transformer/Informer/Autoformer	PV datasets from 2 Australian sites + a northern China PV plant; meteo + PV aggregated to hourly
[12]	Multi-temporal forecasting of wind energy production using artificial intelligence models	1-h/1-day/1-week/1-month ahead	SVM (RBF), ANN, bagged decision trees; bootstrap CIs + Wilcoxon tests; MAE/RMSE/R/R^2/BIAS	Kelmarsh wind farm (UK), SCADA 10-min data; train 2020, test Jan 2021
[40]	WNPS-LSTM-Informer: A Hybrid Stacking model for medium-term photovoltaic power forecasting with ranked feature selection	24 h/96 h/168 h (medium-term)	Proposed: NeuralProphet tuned by GA-enhanced optimizer (WNPS) + ranked feature selection; stacking LSTM+Informer with KNN meta-model	Yulara Solar Site-3A (Uluru, Australia), DKA Solar Centre; PV + meteorological inputs

Table 2 continued

Author	Paper title	Horizon (forecast)	Models used	Context
[21]	Forecasting Wind Farm Production in the Short, Medium, and Long Terms Using Various Machine Learning Algorithms	Hourly/daily/monthly	ANN/MLP, RF, XGBoost, KNN; compares scaling (Min–Max vs Standard)	Söke–Çatalbük wind plant (Turkey), 30 MWe; EPİAŞ production + meteorological data (2018–2022)
[41]	Full-real-life Wind Power Prediction based on Gradient Boosting Decision Tree	15-min, 4 h, 1-day, 4-days	XGBoost and LightGBM (GBDT) with Optuna tuning, 12-fold CV, early stopping	Offshore wind farm (China), 3 sites; 19 months power data + ECMWF HRES 10-day forecasts
[8]	The Effect of Data Pre-processing Techniques on the Accuracy of Medium-Term Solar Power Forecasting	3-day ahead	Linear Regression, Decision Tree Regressor, XGBoost; compares missing data handling + outliers + normalization	PV forecasting on 5-min datasets: Yulara (Uluru) and DKASC Alice Springs (Australia); key finding: row deletion often best
[4]	In-situ PV generation forecasting of utility-scale solar power plants in Ghana: Single and hybrid machine learning approaches	1-day/1-week/2-weeks/1-month	Single: XGBoost, RF, ANN, LR, SVR, Bayesian Regression; hybrids: RF-ANN, RF-XGBoost, XGBoost-ANN	50 MWp Bui PV plant (Ghana); in-situ SCADA 15-min (2021–2022), validation Jan 2023
[2]	Reliability of regression based hybrid machine learning models for the prediction of solar photovoltaics power generation	3-day and 7-day ahead	MARS baseline + residual-learning hybrids (MARS-SVM, MARS-RF, MARS-GBM)	PVWatts/NSRDB-simulated 100 kW PV in North Dakota (3 locations); reliability via error distributions
[37]	A short- and medium-term forecasting model for roof PV systems with data pre-processing	1-hour ahead and 1-day ahead	MLP and LSTM; preprocessing: linear regression for missing weather + k-NN for missing PV + outlier handling + feature selection + scaling	17 rooftop PV sites across Taiwan; Jan 2021–Jun 2023 data (high-frequency, aggregated)
[62]	Developing a Deep Learning and Reliable Optimization Techniques for Solar Photovoltaic Power Prediction	Next-day (30-min intervals)	Hybrid CNN+LSTM (HDNN) optimized with Firefly Algorithm; baselines: LSTM-RNN, MLP, FCNN	Halifax (Nova Scotia, Canada); 1 year 5-min daytime PV data (43,800 samples), 70/30 split
[44]	An intuitive unified hybrid approach for medium to long-term forecasting of different renewable energy sources	Week-ahead/month-ahead/year-ahead	STL-ARIMA-BiLSTM (univariate); benchmarks: ELM, RBFNN, LSTM	India (POSOCO Southern Region) wind/hydro/solar generation series; no explicit weather inputs
[47]	Quantile-transformed multi-attention residual framework (QT-MARF) for medium-term PV and wind power prediction	Next 20 days (tested Jan 1–20)	Quantile Transformer preprocessing + Transformer encoder-decoder + residual blocks	Evaluated on Natal + Santa Vitoria datasets and Chinese State Grid (CSG) PV+wind datasets

Table 2 continued

Author	Paper title	Horizon (forecast)	Models used	Context
[29]	Medium to Long Term Wind Power Prediction Based on Improved mRMR Algorithm and GWO-LSTM	Up to 14-day ahead	Improved mRMR feature selection + GWO-LSTM (Gray Wolf Optimizer tuned LSTM)	Regional wind forecasting (Shandong Province, China) using ECMWF forecast data
[27]	Hybrid Inception-embedded deep neural network ResNet for short and medium-term PV-Wind forecasting	Medium-term ∼25–32 days; short-term ∼24 h	IEDN-RNET: ResNet + Inception + BiLSTM + BiGRU + Time2Vec; baselines: IAMFN, CNN-RNN, CNN-BiLSTM	CSG wind & PV sites (MW-scale) + Natal wind dataset
[55]	Predicting photovoltaic power production using high-uncertainty weather forecasts	1–10 days ahead (reported 1/3/7-day)	SolarPredictor: Residual U-Net + residual composer (causal dilated Conv1D, recurrent connections), trained on forecast ages	SolarDB: 16 PV plants, 1 year; 5-min power + hourly measured + forecast weather with multi-age forecasts
[36]	TFT-Powered PV Production Forecasting: A Multivariate Approach for Long, Medium, and Short Terms	1-year/quarterly/day-ahead	Temporal Fusion Transformer (TFT) with quantile regression; feature filtering via correlation + PCA	Hourly dataset (Greece weather station + "solar power"); multivariate forecasting
[68]	Output of On-Grid Rooftop Photovoltaic Arrays: Case Study for a Sri Lankan Solar Panel Installer	Month-level forecasting	SVR, DT, Ridge/Lasso/LR, XGBoost, Bayesian Ridge, Elastic Net, Polynomial Reg., SGD, ANN, KNN	Installer dataset (Sri Lanka); features emphasize PV system + location/roof dynamics; includes scaling/outlier handling
[42]	A Novel Hybrid Model for Medium-Term Wind Power Forecast Using WDlinear, LightGBM, and Kalman Filtering	7-day ahead	WDlinear + LightGBM (0.5/0.5 weighting) + Kalman filter post-processing; baselines: ARIMA, LSTM, Transformer-family	Two wind farm datasets (1 year each, 15-min); multi-height wind features
[66]	Accurate solar PV power prediction interval method based on frequency-domain decomposition and LSTM model	1-day/7-day/14-day ahead	EEMD decomposition + LSTM (Bayesian optimization) + SVR (zero–one sequence); reported as LSTM-SVR-BO approach	Two Arizona utility-scale PV plants (100 MW, 110 MW), 5-min power data (NREL-style)

Table 2 continued

Author	Paper title	Horizon (forecast)	Models used	Context
[6]	Deep Learning-Based Short-Term Power Output Prediction using Hybrid CNN-LSTM Model for Calatagan Solar Farm, Philippines	3-day and 7-day ahead	Hybrid CNN–LSTM (1D CNN + max-pooling + stacked LSTM + dropout + FC; Adam)	Calatagan Solar Farm case; uses NREL PVWatts simulated inputs to generate PV output
[22]	Exploring Forecasting and Prediction Processes for Decision-Making to Promote the Photovoltaic Energy Integration into the Grid: A Mini Review	All horizons (very short to long term)	Review: persistence, physical/NWP, statistical (ARIMA/SARIMA), ML/DL (ANN/MLP/SVM/RF/boosting), hybrid; metrics and decision aspects	Conceptual/mini-review for PV forecasting and grid integration (no new empirical dataset)
[1]	Computationally expedient Photovoltaic power Forecasting: A LSTM adaptive ensemble method	5-min to 14-days ahead	Ensemble of LSTM models for different horizons + adaptive weighting (RMSE-rank weighting)	DKASC Yulara Solar System (Australia), 5-min data (2017–2020); focus on efficient multi-horizon forecasting
[45]	Investigating the effect of training time for machine learning-based photovoltaic power forecasting	0–24 h and 24–48 h	ExtraTrees Regressor, XGBoost, LightGBM, Linear Regression; main focus on effect of training length	2-year Hungarian PV plant dataset + NWP weather forecasts + solar angles
[26]	Machine Learning-Based Digital Twin for Predictive Modeling in Wind Turbines	1–4 weeks ahead	TCN for wind speed forecasting + k-NN regression (power); grid-search for optimal k	Onshore wind farm (Yalova, Turkey), SCADA 10-min; digital twin framework
[3]	Solar PV Power Forecasting Using Traditional Methods and Machine Learning Techniques	1-day and 1-week ahead	CNN, multi-headed CNN, CNN-LSTM; traditional baselines: ARMA and Multiple Linear Regression (MLR)	UQ Solar (University of Queensland) PV dataset; 15-min meteorological inputs
[35]	Operationalizing Solar Energy Predictions for Sustainable, Autonomous IoT Device Management	3 h to 3 days ahead	Random Forest Regressor (main/best), ANN (1 hidden layer), DNN (3 hidden layers); baselines: EWMA	Solar energy intake prediction for solar-powered IoT devices to support autonomous management
[65]	Wind energy prediction modeling using LSTM-GA networks	Medium-term ~1 month	LSTM + Genetic Algorithm (GA) to optimize time-window / lag size; evaluation via RMSE (+ t-test vs standard LSTM)	Time-series wind measurement dataset (multi-year), 80/20 split while keeping temporal order
[19]	Advances in Wind Power Forecasting (book chapter)	All horizons (conceptual categorization)	Review chapter (no new model): persistence, physical (NWP/CFD), statistical (e.g., ARIMA), ML, hybrid; evaluation metrics	Book chapter overview of wind power forecasting motivation, challenges, time scales, and assessment methods

Table 3 Screening log of excluded studies and exclusion reasons (final screening outcome: 38 excluded, 31 included)

Author	Paper title	Title OK?	Abstract OK?	Exclusion reason
[74]	A Deep Learning Method for Photovoltaic Power Generation Forecasting Based on a Time-Series Dense Encoder	Yes	No	Maximum forecast horizon: 1 day
[53]	Short- and medium-term horizon photovoltaic power forecasting based on terrestrial solar radiation features	Yes	No	Maximum horizon: 24 hours
[13] (2025)	Short-Medium-Term Solar Irradiance Forecasting with a CEEMDAN-CNN-ATT-LSTM Hybrid Model Using Meteorological Data	Yes	No	Maximum forecast horizon: 12 hours
[48]	Forecasting Solar PV Panel Performance Using Linear Regression and Stepwise Linear Regression Machine Learning Algorithms	Yes	Yes/No	Output prediction/estimation from sensor inputs (tilt/irradiance/etc.) rather than a clearly defined forecasting horizon
[49]	Predicting Power Output of Solar Photovoltaic Panels Using Machine Learning Techniques	Yes	Yes/No	Power output prediction/estimation from measured inputs (incl. tilt angle), no defined forecasting horizon
[70]	Machine Learning-Based Prediction Model in Digital Twin System of Wind Farm	Yes	Yes	Forecasts wind speed (not wind power/energy production)
[33]	Impact of Dataset Size on Machine Learning Regression Accuracy in Solar Power Prediction	Yes/No	No	No clear "X hours/days ahead" forecasting task; focuses on training-set size for regression accuracy
[54]	Long-Term Photovoltaic Power Forecasting with Transformer NN	Yes	Yes	Full paper not accessible (Springer access issue)
[14]	Optimizing Wind Turbine Reliability through a Machine Learning Predictive Maintenance Framework	Yes/No	No	Predictive maintenance / fault forecasting, not generation forecasting
[30]	Harnessing AI for solar energy: Emergence of transformer models	Yes	No	Review of methods; no empirical forecasting study
[11]	Towards Automated Model Selection for Wind Speed and Solar Irradiance Forecasting	Yes	No	Maximum horizon: 24 hours
[67]	Medium-term forecasting of global horizontal solar radiation in Brazil using machine learning-based methods	No	No	Forecasts irradiance (GHI) instead of production
[39]	Long, short, and medium terms wind speed prediction model based on LSTM optimized by improved moth flame optimization algorithm	No	No	Wind speed forecasting, not power production
[5]	A hybrid machine-learning model for solar irradiance forecasting	No	No	Solar irradiance forecasting, not power production
[69]	Short-Term Photovoltaic Output Prediction Based on Decomposition and Reconstruction and XGBoost under Two Base Learners	No	No	Short-term forecasting focus
[23]	MSTL-NNAR: a new hybrid model of machine learning and time series decomposition for wind speed forecasting	No	No	Wind speed forecasting, not power production

Table 3 continued

Author	Paper title	Title OK?	Abstract OK?	Exclusion reason
[73]	Optimal Short-term Scheduling of Multiple Small-scale Hydropower Generators based on Water Inflow Prediction	No	No	Hydropower focus, not wind/solar production forecasting
[57]	Performance analysis of machine learning algorithms for hybrid power generation prediction	Yes	No	Prediction/regression without a clear "X hours/days ahead" forecasting setup
[72]	Forecasting Wind Energy Production: Analysis of Meteorological and Temporal Variables Using Optimized Regression Modeling	Yes	Yes/No	No clear forecasting setup; closer to regression on hourly data
[59]	Comparison of Echo State Network with ANN-Based Forecasting Model for Solar Power Generation Forecasting	Yes	Yes	Full paper not accessible (Springer access issue)
[34]	Intelligent Power and Cost Management System in Small-Scale Grid-Associated PV-Wind Energy System Using RNN-LSTM	Yes	No	Smart-grid energy management/load analysis; not multi-day production forecasting
[16]	Cloud cover bias correction in numerical weather models for solar energy monitoring and forecasting systems with kernel ridge regression	No	No	Predicts cloud cover (TCDC), not renewable production
[9]	The Effect of Increasing Solar Photovoltaic Generation on Medium-Term Net-Load Forecasting	No	No	Net-load forecasting, not production forecasting
[18]	Medium-Term Load Forecasting Using ANN and RNN in Microgrid Integrating Renewable Energy Source	No	No	Load demand / conventional generator output forecasting, not renewable production
[63]	Time Load Forecasting: A Smarter Expertise Through Modern Methods	No	No	Load demand forecasting
[20]	Improved Hourly Prediction of BIPV Photovoltaic Power Building Using Artificial Learning Machine: A Case Study	No	No	Hourly prediction focus (not medium-term multi-day)
[64]	Transient heat transfer performance prediction using a machine learning approach for sensible heat storage in parabolic trough solar thermal power generation cycles	No	No	Thermal storage performance simulation, not renewable production forecasting
[56]	Medium-term forecasting of power generation by hydropower plants in isolated power systems under climate change	No	No	Hydropower focus
[17]	Forecasting solar photosynthetic photon flux density under cloud cover effects: novel predictive model using CNN-LSTM	Yes/No	No	Very-short-term solar radiation forecasting (5-minute horizon)
[43]	Deep Recurrent Neural Networks for Building-Level Load Forecasting	No	No	Load forecasting

Table 3 continued

Author	Paper title	Title OK?	Abstract OK?	Exclusion reason
[10]	One-Day-Ahead Solar Irradiation and Windspeed Forecasting with Advanced Deep Learning Techniques	No	No	Irradiation and wind speed forecasting (not production)
[61]	Medium-term regional electricity load forecasting through machine learning and deep learning	No	No	Load forecasting
[50]	Medium-Term Wind Speed Prediction using Bayesian Neural Network (BNN)	No	No	Wind speed prediction, not power production
[31]	Development of high-precision wind, wave and current forecast system for offshore wind energy industry in Taiwan: a two-stage method of numerical simulation and AI correction	No	No	Meteorological field forecasting; not power generation forecasting
[46]	Very-Short Term Electricity Demand Forecasting Integrating Solar Distributed Generation	No	No	Short-term demand forecasting
[60]	Ultra-short-term prediction method of photovoltaic electric field power based on ground-based cloud image segmentation	No	No	Short-term forecasting
[15]	Mixed Regression Clustering Techniques for the Medium Term Prediction of Reactive Power within Transmission Grids	No	No	Reactive power consumption forecasting
[32]	Research on Predicting the Short-term Output of Photovoltaic (PV) Based on Extreme Learning Machine Model and Improved Similar Day	No	No	Short-term forecasting

Author Contributions Dani Mahaini was contributed conceptualization, methods, data collection, analysis, software, and writing draft; Marcos R. Machado was involved in conceptualization, validation, project administration, supervision, and writing draft/review; and João Luiz Rebelo Moreira was done conceptualization, supervision, and writing review.

Data Availability No datasets were generated or analyzed during the current study.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Ahmed, R., Sreeram, V., Togneri, R., Datta, A., Arif, M.: Computationally expedient photovoltaic power forecasting: A lstm ensemble method augmented with adaptive weighting and data segmentation technique. Energy Conversion and Management, *258*. https://www.scopus.com/inward/record.uri?eid=2-s2.0-85127327484&doi=10.1016%2fj.enconman.2022.115563&partnerID=40&md5=62b6f258048d3da49678648b4407580a. https://doi.org/10.1016/j.enconman.2022.115563. Cited By 68 (2022)
2. Ahmed, S., Bhuiyan, K., Rahman, I., Salehfar, H., Selvaraj, D.: Reliability of regression based hybrid machine learning models for the prediction of solar photovoltaics power generation. Energy Reports, *12*, 5009–5023. https://www.scopus.com/inward/record.uri?eid=2-s2.0-85208397144&doi=10.1016%2fj.egyr.2024.10.060&partnerID=40&md5=6bd759430d528d87ea5ef8415df9fb18. https://doi.org/10.1016/j.egyr.2024.10.060. Cited By 7 (2024)
3. Alam, A., Nahid-Al-Masood, Iqbal Asif Razee, M., Zunaed, M.: Solar pv power forecasting using traditional methods and machine learning techniques. (2021). https://www.scopus.com/inward/record.uri?eid=2-s2.0-85109394656&doi=10.1109%2fKPEC51835.2021.9446199&partnerID=40&md5=7758e147862a60d64ee2fdabea894f17. https://doi.org/10.1109/KPEC51835.2021.9446199 cited By 29
4. Alhassan, M., Gyamfi, S., Aboagye, B., Diawuo, F., Kwarteng, M.: In-situ pv generation forecasting of utility-scale solar power plants in ghana: Single and hybrid machine learning approaches. Solar Energy Advances, *5* (2025) https://www.scopus.com/inward/record.uri?eid=2-s2.0-105022817793&doi=10.1016%2fj.seja.2025.100124&partnerID=40&md5=141592eabf6f56d5ce25a486c4b2f8b6. https://doi.org/10.1016/j.seja.2025.100124. Cited By 0
5. Almarzooqi, A.M., Maalouf, M., El-Fouly, T.H., Katzourakis, V.E., El Moursi, M.S., Chrysikopoulos, C.V.: A hybrid machine-learning model for solar irradiance forecasting. Clean Energy **8**, 100–110 (2024)
6. Andal, J., Aytona, F., Duque, M., Marquez, J., Villaflor, P., Salvador, G., Tubola, O.: Deep learning-based short-term power output prediction using hybrid cnn-lstm model for calatagan solar farm, philippines. (2023) (pp. 364–369). https://www.scopus.com/inward/record.uri?eid=2-s2.0-86000005975&doi=10.1109%2fAIRoSIP58759.2023.10873867&partnerID=40&md5=49814a683eabc85ba3c35d09745e28e5. https://doi.org/10.1109/AIRoSIP58759.2023.10873867 cited By 1
7. Aydin, E., Onal, K., Haydaroglu, C., Kiliç, H., Yildirim, O., Katar, O., Erdoğan, H.: A novel scenario-based comparative framework for short- and medium-term solar pv power forecasting using deep learning models. *Applied Sciences Switzerland*, *15* (2025). https://www.scopus.com/inward/record.uri?eid=2-s2.0-105025969685&doi=10.3390%2fapp152412965&partnerID=40&md5=a4b22b2590417e79dc5511a4ab10a2ce. https://doi.org/10.3390/app152412965. Cited By 0
8. Bergman, S., Wang, B., Weng, Y., Zhang, Y., Liu, Y., Ma, X.: The effect of data pre-processing techniques on the accuracy of medium-term solar power forecasting. (pp. 144–148) (2025) https://www.scopus.com/inward/record.uri?eid=2-s2.0-105017771207&doi=10.1109%2fInECCE64959.2025.11150818&partnerID=40&md5=af08ba5a46869d64eff6a9793cc714e5. https://doi.org/10.1109/InECCE64959.2025.11150818 cited By 0
9. Blain, J., Donaldson, D. L.: The effect of increasing solar photovoltaic generation on medium-term net-load forecasting. In *Renewable Power Generation and Future Power Systems Conference 2023 (RPG 2023 UK)* (pp. 94–100). IET volume 2023 (2023)
10. Blazakis, K., Katsigiannis, Y., Stavrakakis, G.: One-day-ahead solar irradiation and windspeed forecasting with advanced deep learning techniques. Energies **15**, 4361 (2022)
11. Blazakis, K., Schetakis, N., Bonfini, P., Stavrakakis, K., Karapidakis, E., Katsigiannis, Y.: Towards automated model selection for wind speed and solar irradiance forecasting. Sensors **24**, 5035 (2024)
12. Bouabdallaoui, D., Haidi, T., Derri, M., Hbiak, I., El Jaadi, M.: Multi-temporal forecasting of wind energy production using artificial intelligence models. *International Journal of Renewable Energy Development*, *14*, 505–517 (2025). https://www.scopus.com/inward/record.uri?eid=2-s2.0-105004822584&doi=10.61435%2fijred.2025.61086&partnerID=40&md5=5ff5d3fa2a44a718ed210ed38522c97f. https://doi.org/10.61435/ijred.2025.61086. Cited By 0
13. Camacho, M., Maldonado-Correa, J., Torres-Cabrera, J., Martín-Martínez, S., Gómez-Lázaro, E.: Short-medium-term solar irradiance forecasting with a ceemdan-cnn-att-lstm hybrid model using meteorological data. Appl. Sci. **15**, 1275 (2025)
14. Cheng, J.C., Bronskiy, S., Cheung, C.K.: Optimizing wind turbine reliability through a machine learning predictive maintenance framework. In: 2025 IEEE International Symposium on the Application of Artificial Intelligence in Electrical Engineering (AAIEE), pp. 281–290. IEEE (2025)
15. Dapoz, P.-D., De Grève, Z., De Pauw, P., Sprooten, J., Vallee, F.: Mixed regression clustering techniques for the medium term prediction of reactive power within transmission grids. In: 2019 IEEE PES Innovative Smart Grid Technologies Europe, pp. 1–5. IEEE (2019)
16. Deo, R.C., Ahmed, A.M., Casillas-Perez, D., Pourmousavi, S.A., Segal, G., Yu, Y., Salcedo-Sanz, S.: Cloud cover bias correction in numerical weather models for solar energy monitoring and forecasting systems with kernel ridge regression. Renewable Energy **203**, 113–130 (2023)
17. Deo, R.C., Grant, R.H., Webb, A., Ghimire, S., Igoe, D.P., Downs, N.J., Al-Musaylh, M.S., Parisi, A.V., Soar, J.: Forecasting solar photosynthetic photon flux density under cloud cover effects: novel predictive model using convolutional neural network integrated with long short-term memory network. Stoch. Env. Res. Risk Assess. **36**, 3183–3220 (2022)

18. Dewangan, F., Biswal, M.: Medium-term load forecasting using ann and rnn in microgrid integrating renewable energy source. In: 2023 2nd International Conference for Innovation in Technology (INOCON), pp. 1–5. IEEE (2023)
19. Dittner, M., Vasel, A.: Advances in wind power forecasting. *Lecture Notes in Energy*, *70*, 37–57 (2019). https://www.scopus.com/inward/record.uri?eid=2-s2.0-85064623602&doi=10.1007%2f978-3-030-05636-0_3&partnerID=40&md5=716c846ed75f9b994b6cb33c38bb6cef. https://doi.org/10.1007/978-3-030-05636-0_3. Cited By 2
20. Dourhmi, M., Benlamine, K., Abouelaziz, I., Zghal, M., Masrour, T., J., Y.: Improved hourly prediction of bipv photovoltaic power building using artificial learning machine: a case study. In: International Conference on Networking, Intelligent Systems and Security, pp. 270–280. Springer (2022)
21. Ekinci, G., Ozturk, H.: Forecasting wind farm production in the short, medium, and long terms using various machine learning algorithms. *Energies*, *18* (2025). https://www.scopus.com/inward/record.uri?eid=2-s2.0-86000644762&doi=10.3390%2fen18051125&partnerID=40&md5=c357b9e4c7b314d2d7e4b8993e72cdf1. https://doi.org/10.3390/en18051125. Cited By 5
22. El Robrini, F., Amrouche, B.: Exploring forecasting and prediction processes for decision-making to promote the photovoltaic energy integration into the grid: A mini review. (pp. 180–186). (2023). https://www.scopus.com/inward/record.uri?eid=2-s2.0-85177439774&doi=10.1109%2fDASA59624.2023.10286623&partnerID=40&md5=55ba5d44fdf2b48ed848c56674c3be10. https://doi.org/10.1109/DASA59624.2023.10286623 cited By 5
23. Elseidi, M.: Mstl-nnar: a new hybrid model of machine learning and time series decomposition for wind speed forecasting. Stoch. Env. Res. Risk Assess. **38**, 2613–2632 (2024)
24. EN2014, W.: The cost of wind power forecast errors in the belgian power system, (2014)
25. Eurostat 2024: nearly 50% of eu electricity came from renewables. Eurostat. https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20260114-1? accessed: 2026-02-19 (2026)
26. Fahim, M., Sharma, V., Cao, T.-V., Canberk, B., Duong, T.: Machine learning-based digital twin for predictive modeling in wind turbines. *IEEE Access*, *10*, 14184–14194 (2022) https://www.scopus.com/inward/record.uri?eid=2-s2.0-85124066585&doi=10.1109%2fACCESS.2022.3147602&partnerID=40&md5=d710aa74766f65139915d70a061a6ff4. https://doi.org/10.1109/ACCESS.2022.3147602. Cited By 154
27. Feroz Mirza, A., Mansoor, M., Usman, M., Ling, Q.: Hybrid inception-embedded deep neural network resnet for short and medium-term pv-wind forecasting. *Energy Conversion and Management*, *294* (2023) https://www.scopus.com/inward/record.uri?eid=2-s2.0-85169566054&doi=10.1016%2fj.enconman.2023.117574&partnerID=40&md5=5ce64c2bdb372220ff5b7c026a7a7737. https://doi.org/10.1016/j.enconman.2023.117574. Cited By 62
28. Fraccascia, L., Yazdanpanah, V., Van Capelleveen, G., Yazan, D.M.: Energy-based industrial symbiosis: a literature review for circular energy transition. Environ. Dev. Sustain. **23**, 4791–4825 (2021)
29. Guo, Z., Zhao, F., Wang, B., Xu, W., Zhang, T., Guo, K.: Medium to long term wind power prediction based on improved mrmr algorithm and gwo-lstm algorithm. (pp. 137–141) (2024) https://www.scopus.com/inward/record.uri?eid=2-s2.0-85204405572&doi=10.1109%2fICESEP62218.2024.10651978&partnerID=40&md5=1f0f91ed5a579b28296b32273ae82b7c. https://doi.org/10.1109/ICESEP62218.2024.10651978 cited By 4
30. Hanif, M., Mi, J.: Harnessing ai for solar energy: Emergence of transformer models. Appl. Energy **369**, 123541 (2024)
31. Hung, S.-C., Chang, W.-Y., Tsai, W.-F., Kuo, C.-Y., Liau, J.-M., Lin, C.-Y., Hsu, T.-W., Tu, C.-Y., Wu, C.-H., Chung, Y.-F., et al.: Development of high-precision wind, wave and current forecast system for offshore wind energy industry in taiwan: a two-stage method of numerical simulation and ai correction. J. Chin. Inst. Eng. **44**, 532–543 (2021)
32. Jiang, Y., Yang, Y., Wu, Q., Chi, X., Miao, J.: Research on predicting the short-term output of photovoltaic (pv) based on extreme learning machine model and improved similar day. In *2019 IEEE Innovative Smart Grid Technologies-Asia (ISGT Asia)* (pp. 3691–3695) (2019) IEEE
33. Karim, S., Hossain, M.S., Akter, K., Sarker, D., Kabir, M.M., A., M.: Impact of dataset size on machine learning regression accuracy in solar power prediction, p. 122. Energy Engineering (2025)
34. Kothai Andal, C., Jayapal, R.: Intelligent power and cost management system in small-scale grid-associated pv-wind energy system using rnn-lstm. SN Computer Science **4**, 746 (2023)
35. Kraemer, F., Palma, D., Braten, A., Ammar, D.: Operationalizing solar energy predictions for sustainable, autonomous iot device management. *IEEE Internet of Things Journal*, *7*, 11803–11814 (2020) https://www.scopus.com/inward/record.uri?eid=2-s2.0-85097831928&doi=10.1109%2fJIOT.2020.3002330&partnerID=40&md5=412d15542b236eb213db9dbc57358019. https://doi.org/10.1109/JIOT.2020.3002330. Cited By 33
36. Kumar, D., Picerno, S.: Tft-powered pv production forecasting: A multivariate approach for long, medium, and short terms (2023) https://www.scopus.com/inward/record.uri?eid=2-s2.0-85187800174&doi=10.1109%2fCERA59325.2023.10455743&partnerID=40&md5=e33f22b8c24d6463b8f5bf2a9606ad20. https://doi.org/10.1109/CERA59325.2023.10455743 cited By 2
37. Lee, D.-S., Lai, C.-W., Fu, S.-K.: A short- and medium-term forecasting model for roof pv systems with data pre-processing. *Heliyon*, *10* (2024). https://www.scopus.com/inward/record.uri?eid=2-s2.0-85188511456&doi=10.1016%2fj.heliyon.2024.e27752&partnerID=40&md5=328823f5cfafde63b31a12bb39eb3180. https://doi.org/10.1016/j.heliyon.2024.e27752. Cited By 10
38. Li, J., Ren, X., Zhang, F., Gao, L., Li, J., Cui, T.: Multi-timescale photovoltaic station power prediction based on reformer model. *Energy*, *327* (2025a) https://www.scopus.com/inward/record.uri?eid=2-s2.0-105004414910&doi=10.1016%2fj.energy.2025.136423&partnerID=40&md5=1637b815f83eb0314c7ca8e840082b71. https://doi.org/10.1016/j.energy.2025.136423. Cited By 2
39. Li, R., Wang, J., Li, J., Kou, M.: Long, short, and medium terms wind speed prediction model based on lstm optimized by improved moth flame optimization algorithm. Environ. Sci. Pollut. Res. **31**, 37256–37282 (2024)
40. Li, Y., Liu, G., Cao, Y., Chen, J., Gang, X., Tang, J.: Wnps-lstm-informer: A hybrid stacking model for medium-term photovoltaic power forecasting with ranked feature selection. Renewable Energy, *244* (2025b) https://www.scopus.com/inward/record.uri?eid=2-s2.0-85218883321&doi=10.1016%2fj.renene.2025.122687&partnerID=40&md5=36de40f63ac88194d0b26b8a83d3d207. https://doi.org/10.1016/j.renene.2025.122687. Cited By 16
41. Li, Y., Qi, W., Zhou, S.: Full-real-life wind power prediction based on gradient boosting decision tree. IEEE Information Technology and Mechatronics Engineering Conference Itoec, (pp. 193–198) (2025c) https://www.scopus.com/inward/record.uri?eid=2-s2.0-105005835487&doi=10.1109%2fITOEC63606.2025.10968639&partnerID=40&md5=dcf403bf3ee2df6899355ec676a8aa21. https://doi.org/10.1109/ITOEC63606.2025.10968639. Cited By 0
42. Liao, Y., Li, C., Diao, R., Sun, K., Liu, J., Zhu, L., Jiang, Y., Chen, L.: A novel hybrid model for medium-term wind power forecast using wdlinear, lightgbm, and kalman filtering. (pp. 1969–1974)

(2023) https://www.scopus.com/inward/record.uri?eid=2-s2.0-85194131092&doi=10.1109%2fEI259745.2023.10512357&partnerID=40&md5=d2d4ea07e02fe7e33986a149717ee78c. https://doi.org/10.1109/EI259745.2023.10512357 cited By 1
43. Linaro, D., Del Giudice, D., Bizzarri, F., Brambilla, A.: Deep recurrent neural networks for building-level load forecasting. In: 2022 IEEE International Symposium on Circuits and Systems (ISCAS), pp. 118–122. IEEE (2022)
44. Malhan, P., Mittal, M.: An intuitive unified hybrid approach for medium to long-term forecasting of different renewable energy sources. International Journal of Ambient Energy, *45* (2024). https://www.scopus.com/inward/record.uri?eid=2-s2.0-85177599790&doi=10.1080%2f01430750.2023.2267554&partnerID=40&md5=78407d349b000f44d3ef41493c3697f6. https://doi.org/10.1080/01430750.2023.2267554. Cited By 0
45. Markovics, D., Mayer, M.: Investigating the effect of training time for machine learning based photovoltaic power forecasting (2022) https://www.scopus.com/inward/record.uri?eid=2-s2.0-85137799051&doi=10.1109%2fIYCE54153.2022.9857544&partnerID=40&md5=cd70df7e5c6bdc6a64166a1e949a335d. https://doi.org/10.1109/IYCE54153.2022.9857544 cited By 4
46. Mejía-Giraldo, D., Casadiegos-Osorio, A., Grajales-Espinal, C., Tobón-Villa, J. E.: Very-short term electricity demand forecasting integrating solar distributed generation. In 2021 IEEE International Conference on Engineering Veracruz (ICEV) (pp. 1–6). (2021) IEEE
47. Mirza, A., Shu, Z., Usman, M., Mansoor, M., Ling, Q.: Quantile-transformed multi-attention residual framework (qt-marf) for medium-term pv and wind power prediction. Renewable Energy, *220* (2024). https://www.scopus.com/inward/record.uri?eid=2-s2.0-85176313254&doi=10.1016%2fj.renene.2023.119604&partnerID=40&md5=dc54abc464b9dcae58f5db46e6241159. https://doi.org/10.1016/j.renene.2023.119604. Cited By 30
48. Mohammad, K.S., Yousuf, A.H., Boddu, M.K., Sam, J.K.M., Chandrashekar, M., Alias, L.: Forecasting solar pv panel performance using linear regression and stepwise linear regression machine learning algorithms. Journal Européen des Systèmes Automatisés **58**, 365–371 (2025)
49. Mohammad, K. S., Yousuf, A. H., Boddu, M. K., Syed, M. S., Ahmad, A. et al.: Predicting power output of solar photovoltaic panels using machine learning techniques. In 2025 International Conference for Artificial Intelligence, Applications, Innovation and Ethics (AI2E) (pp. 1–6) (2025b). IEEE
50. Mohsin, S., Ramli, S.N., Imdad, M.: Medium-term wind speed prediction using bayesian neural network (bnn). Int. J. Syst. Innov **6**, 11–20 (2021)
51. Osifeko, M., Munda, J.: Scenario-based stochastic optimization for renewable integration under forecast uncertainty: A south african power system case study. Processes, *13* (2025). https://www.scopus.com/inward/record.uri?eid=2-s2.0-105014325659&doi=10.3390%2fpr13082560&partnerID=40&md5=838c33ccada7492fffb7f9466a25ce56. https://doi.org/10.3390/pr13082560. Cited By 1
52. Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E. et al.: The prisma 2020 statement: an updated guideline for reporting systematic reviews. bmj, *372* (2021)
53. Parczyk, P., Burduk, R.: Short-and medium-term horizon photovoltaic power forecasting based on terrestrial solar radiation features. Eng. Appl. Artif. Intell. **162**, 112668 (2025)
54. Piantadosi, G., Ferlito, S., Dutto, S., Francia, D., G.: Long-term photovoltaic power forecasting with transformer nn. In: AISEM Annual Conference on Sensors and Microsystems, pp. 250–256. Springer (2024)
55. Polasek, T., Čadík, M.: Predicting photovoltaic power production using high-uncertainty weather forecasts. Applied Energy, *339* (2023) https://www.scopus.com/inward/record.uri?eid=2-s2.0-85151456461&doi=10.1016%2fj.apenergy.2023.120989&partnerID=40&md5=bf23241047c06d8454e58909549bae20. https://doi.org/10.1016/j.apenergy.2023.120989. Cited By 60
56. Safaraliev, M., Kiryanova, N., Matrenin, P., Dmitriev, S., Kokin, S., Kamalov, F.: Medium-term forecasting of power generation by hydropower plants in isolated power systems under climate change. Energy Rep. **8**, 765–774 (2022)
57. Sarışık, G., Öğütlü, A.S.: Performance analysis of machine learning algorithms for hybrid power generation prediction. Neural Comput. Appl. **37**, 3203–3225 (2025)
58. Seman, L., Stefenon, S., Yow, K.-C., Coelho, L., Mariani, V.: Multi-step short-term solar energy forecasting using fourier-enhanced bilstm and neural additive models. Renewable Energy, *257* (2026). https://www.scopus.com/inward/record.uri?eid=2-s2.0-105021263072&doi=10.1016%2fj.renene.2025.124738&partnerID=40&md5=45453183761f27873148d8de15e8ca76. https://doi.org/10.1016/j.renene.2025.124738. Cited By 0
59. Shashikant, S., Nayak, B., J.R.: Comparison of echo state network with ann-based forecasting model for solar power generation forecasting. In: International Conference on Computational Intelligence in Machine Learning, pp. 133–144. Springer (2022)
60. Shen, R., Xing, R., Wang, Y., Hua, D., Ma, M.: Ultra-short-term prediction method of photovoltaic electric field power based on ground-based cloud image segmentation. In E3S Web of Conferences (p. 01052). EDP Sciences volume 185 (2020)
61. Shirzadi, N., Nizami, A., Khazen, M., Nik-Bakht, M.: Medium-term regional electricity load forecasting through machine learning and deep learning. Designs **5**, 27 (2021)
62. Shivashankaraiah, K., Shashibhushan, G., Prema Kirubakaran, A., Anjanappa, C., Sharma, P., Mayiladuthurai Vaidyanathan, I., Taqui, S., Obaid, S., Alfarraj, S.: Developing a deep learning and reliable optimization techniques for solar photovoltaic power prediction. *Electric Power Components and Systems*,. https://www.scopus.com/inward/record.uri?eid=2-s2.0-85186857864&doi=10.1080%2f15325008.2024.2317369&partnerID=40&md5=1e8aa62b0e0b62cd47329344aa1821ec. https://doi.org/10.1080/15325008.2024.2317369. Cited By 7 (2024)
63. Som, T.: Time load forecasting: A smarter expertise through modern methods. In: Prediction Techniques for Renewable Energy Generation and Load Demand Forecasting, pp. 153–176. Springer (2023)
64. Suwa, T.: Transient heat transfer performance prediction using a machine learning approach for sensible heat storage in parabolic trough solar thermal power generation cycles. Journal of Energy Storage **56**, 105965 (2022)
65. Vaitheeswaran, S., Ventrapragada, V.: Wind power pattern prediction in time series measuremnt data for wind energy prediction modelling using lstm-ga networks (2019). https://www.scopus.com/inward/record.uri?eid=2-s2.0-85078086069&doi=10.1109%2fICCCNT45670.2019.8944827&partnerID=40&md5=58095eb5143769041de9d21056736a6d. https://doi.org/10.1109/ICCCNT45670.2019.8944827 cited By 15
66. Wang, L., Mao, M., Xie, J., Liao, Z., Zhang, H., Li, H.: Accurate solar pv power prediction interval method based on frequency-domain decomposition and lstm model. Energy, *262* (2023) https://www.scopus.com/inward/record.uri?eid=2-s2.0-85139325249&doi=10.1016%2fj.energy.2022.125592&partnerID=40&md5=28901cb749d128591995e5866b579ca6. https://doi.org/10.1016/j.energy.2022.125592. Cited By 142
67. Weyll, A.L.C., Kitagawa, Y.K.L., Araujo, M.L.S., da Silva Ramos, D.N., de Lima, F.J.L., dos Santos, T.S., Jacondino, W.D., Silva, A.R., Araújo, A.C., Pereira, L.K.M., et al.: Medium-term forecasting of global horizontal solar radiation in brazil using machine learning-based methods. Energy **300**, 131549 (2024)

68. Wickramasinghe, B., Asanka, P.: Forecasting of medium-term energy output of on-grid rooftop photovoltaic arrays -case study for a sri lankan solar panel installer. (2023) https://www.scopus.com/inward/record.uri?eid=2-s2.0-85170094133&doi=10.1109%2fSCSE59836.2023.10214986&partnerID=40&md5=31945c982027f150473fd02de79f74f4. https://doi.org/10.1109/SCSE59836.2023.10214986 cited By 2
69. Xu, W., Wang, Z., Wang, W., Zhao, J., Wang, M., Wang, Q.: Short-term photovoltaic output prediction based on decomposition and reconstruction and xgboost under two base learners. Energies **17**, 906 (2024)
70. Yang, H., Jiao, Y., Wang, X., Luo, X.: Machine learning-based prediction model in digital twin system of wind farm. In: 2025 37th Chinese Control and Decision Conference (CCDC), pp. 985–990. IEEE (2025)
71. Yu, Y., Junjan, V., Yazan, D.M., Iacob, M.-E.: A systematic literature review on circular economy implementation in the construction industry: a policy-making perspective. Resour. Conserv. Recycl. **183**, 106359 (2022)
72. Yüksek, G.: Forecasting wind energy production: Analysis of meteorological and temporal variables using optimized regression modeling. In: 2024 Global Energy Conference (GEC), pp. 146–152. IEEE (2024)
73. Zhou, Z., Yue, X., Y., Q.: Optimal short-term scheduling of multiple small-scale hydropower generators based on water inflow prediction. In: 2024 IEEE 8th Conference on Energy Internet and Energy System Integration (EI2), pp. 402–407. IEEE (2024)
74. Zi, X., Liu, F., Liu, M., Wang, Y.: A deep learning method for photovoltaic power generation forecasting based on a time-series dense encoder. Energies **18**, 2434 (2025)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.